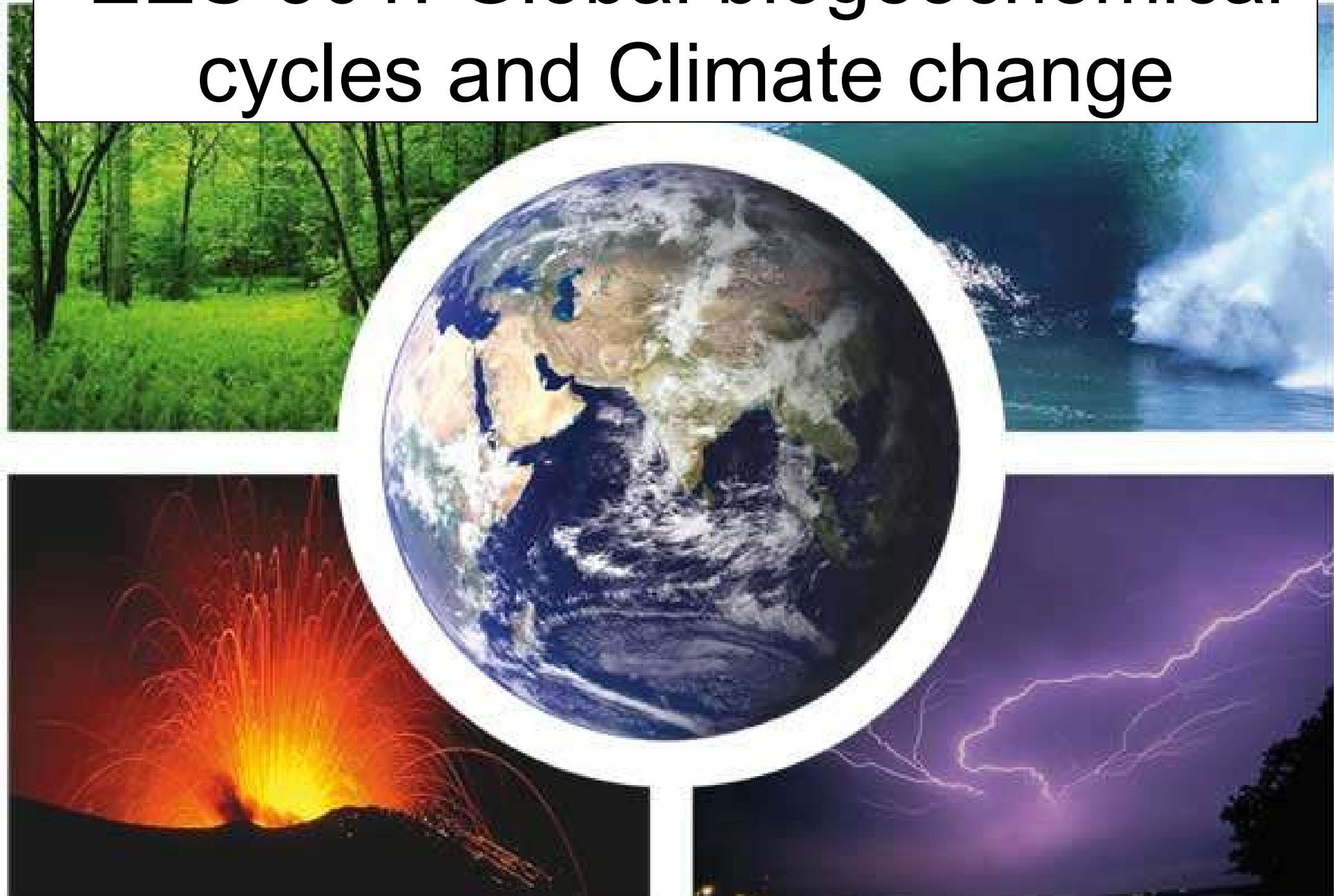


EES 301: Global biogeochemical cycles and Climate change



What is a model

- When we formulate a model, we put all our knowledge about the system into a set of mathematical equations which can
 1. **Form the basis for the interpretation of the past history of the system.**
This means a model can make predictions about the past which we can verify against the geological evidence. Our predictions about when LUCA lived and what genetic capability the organism had are based on genetic models. Carbon isotopic evidence consistent with the existence of certain metabolic processes would form an evaluation.
 2. **Form the basis of predictions of the future of the system**
The International Panel on Climate Change has been modeling future climate starting with simple models and publishes the predictions ca. every 5 years starting with the 1st Assessment report in 1990. For the early models we by now have 35 years of observational climate data to evaluate the model against. Most IPCC reports also contain a section that looks at how well the model does in modeling the quaternary period which contains the relative rapid flips in and out of ice ages as well as the two stable states ice age and warm age.

Models of Biogeochemical cycles

- When we model **biogeochemical cycles** we normally simplify the system into **reservoirs** of relative large spatial extend and look at **monthly or annual averages**.
- This means the day-to-day variations of the weather and ocean currents are not normally considered when modeling Biochemical cycles.
- We have **specialized models such as weather forecast models** when we want to make **detailed spatially resolved predictions about the future weather**, but such models generally don't deal with long term changes in the biogeochemical cycling of elements. Only with the short term changes in the fluxes of some biogeochemical systems (e.g. the hydrological cycle) that matter for the weather forecast.
- We also have **intermediate scale models the so called chemical tracer transport models** which can deal with the spatio-temporally resolved **chemical fluxes of shorter lived species in the carbon and sulfur cycle**. However, these can generally only be run for a couple of months or years.

Single reservoir systems

- The simplest model of a biogeochemical cycle is the single reservoir model. In its simplest form the model has one single source flux Q , one single first order sink flux $S=kM$ and contains the burden of mass M of the substance being modeled.

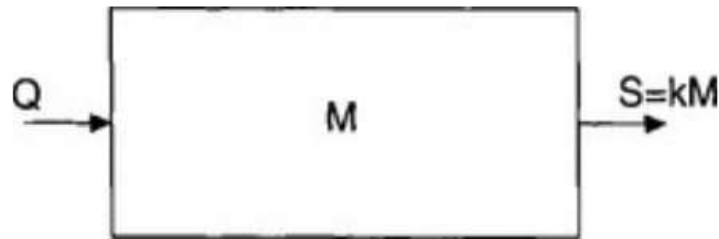


FIG. 4-1 Schematic illustration of a single reservoir with source flux Q , sink flux S , and content M .

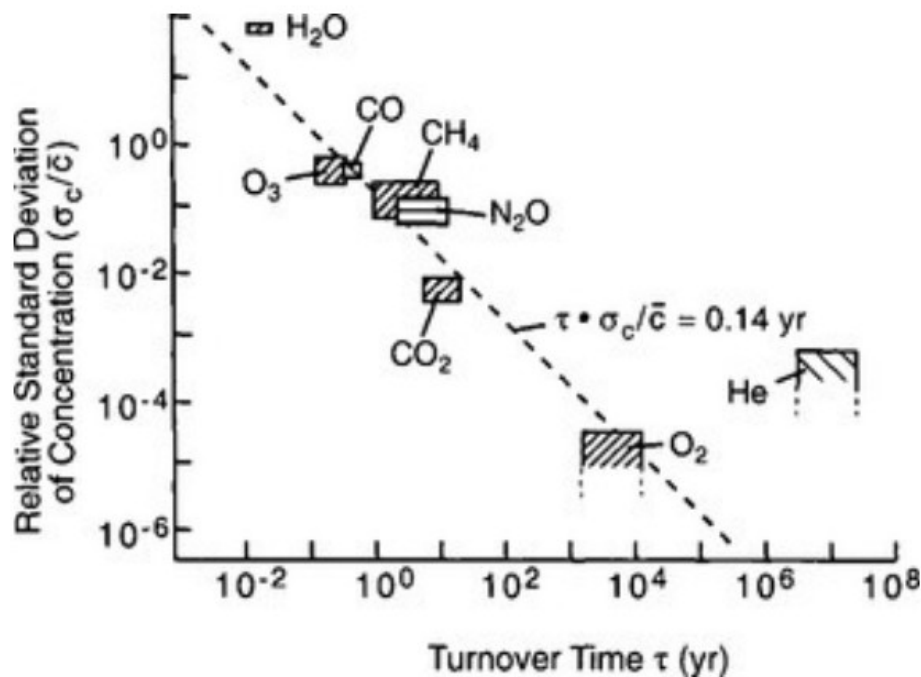
- In this case it is easy to calculate the turnover time defined in lecture two the time it would take to empty the reservoir if the sink S remained constant while sources were zero. Some authors refer to this as the renewal time or flushing time.

$$\tau_0 = \frac{M}{S}$$

$$\tau_0 S = M$$

Turnover time of single reservoir systems

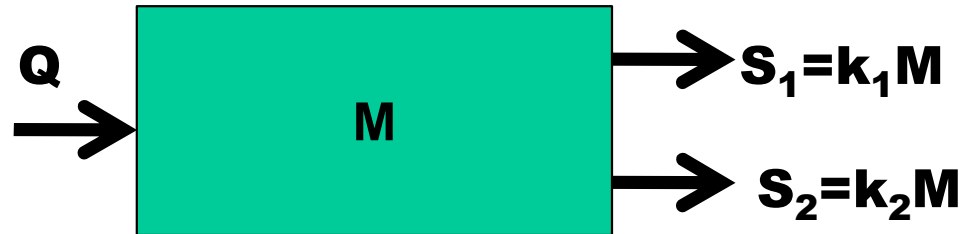
- In common cases where the sink is proportional to the reservoir content $S=kM$ the turnover time is the inverse of the proportionality constant k^{-1} , which is analogous to first-order chemical kinetics.



- In fluid reservoirs like the atmosphere or the ocean the turnover time of a tracer is also related to the spatial and temporal variability of its concentration with the reservoir. A long turnover time corresponds to a small variability and vice versa.
- Deviations from the line are caused by spatially inhomogeneous source strengths

FIG. 4-2 Inverse relationship between relative standard deviation of atmospheric concentration and turnover time for important trace chemicals in the troposphere. (Modified from

Turnover time with more than one sink flux



- When material is removed by two or more processes, each with the flux S_i then the turnover time with respect to each individual flux can be defined as:

$$\tau_{0i} = \frac{M}{S_i} \qquad \tau_{0i} S_i = M$$

While the overall turnover time with respect to the sum of all fluxes is:

$$\sum_i^n \tau_{0i} = \frac{M}{\sum_i^n S_i}$$

Single reservoir systems

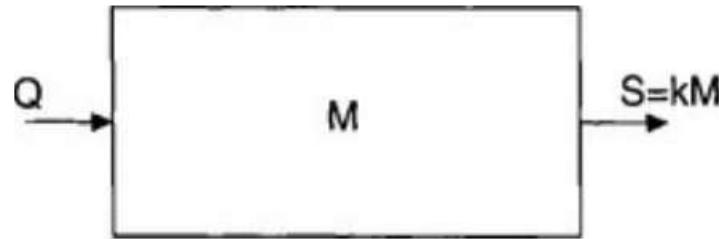


FIG. 4-1 Schematic illustration of a single reservoir with source flux Q , sink flux S , and content M .

- The equation describing the rate of change of the content of a reservoir can be written as:

$$\frac{dM}{dt} = Q - S = Q - \frac{M}{\tau_0}$$

- If the reservoir is in steady state

$$\frac{dM}{dt} = 0$$

then the sources Q and sources S must balance. In steady-state reservoirs Q can replace S in the turnover time equation.

Example of a turnover time calculation with more than one sink flux

Let us consider just one of the reservoirs of the carbon cycle:

Surface biota in the ocean containing a mass of 3×10^{15} g C as burden.

Material is lost through two processes:

- 1) Respiration & decomposition 36×10^{15} g C yr⁻¹
- 2) through detritus falling to the ocean floor 4×10^{15} g C yr⁻¹

The reservoir has a total turnover time of:

$$\sum_i^n \tau_{oi} = \frac{3 \times 10^{15} [g]}{(4 + 36) \times 10^{15} [gyr^{-1}]} = 0.075 [yr] = 0.9 [months]$$

The turnover time is totally dominated by the turnover time with respect to respiration and decomposition $3/36 * 12 =$ exactly 1 month. The turnover time with respect to settling of detritus is 9 months

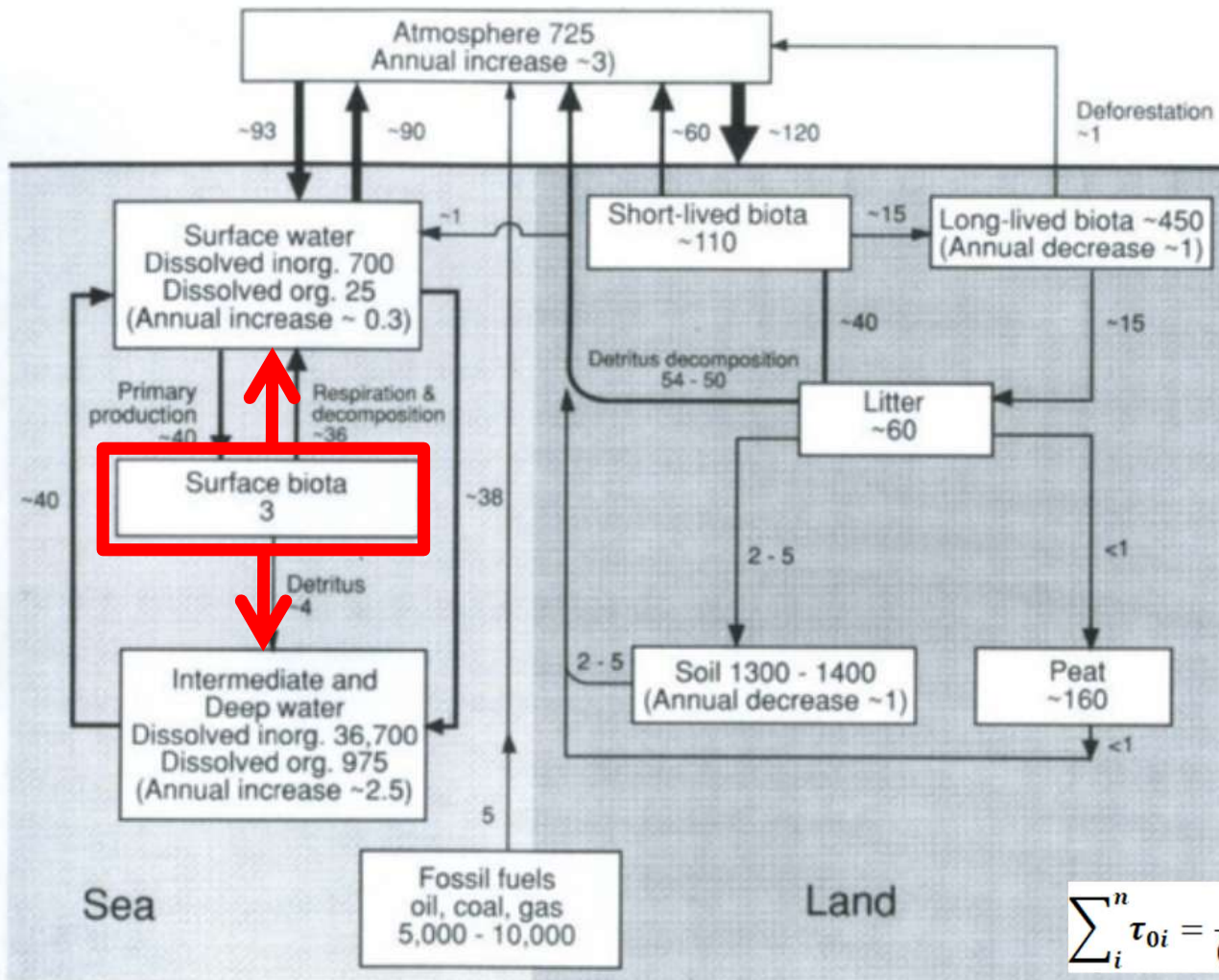


FIG. 4-3 Principal reservoirs and fluxes in the carbon cycle. Units are 10^{15} g (Pg) C (burdens) and Pg C/yr (fluxes). (From [Bolin \(1986\)](#)) with permission from John Wiley and Sons.)

Residence time or transit time

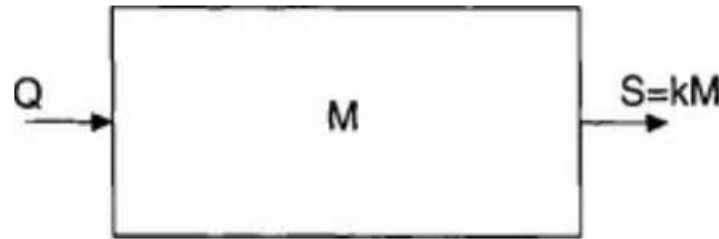


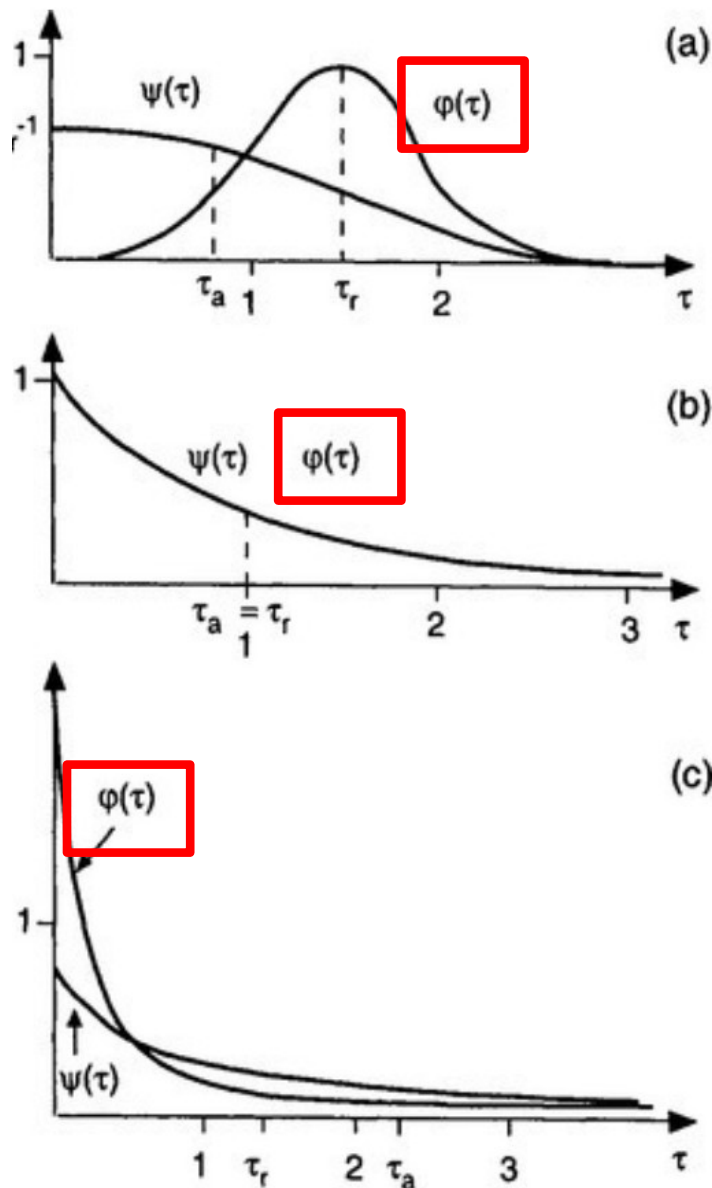
FIG. 4-1 Schematic illustration of a single reservoir with source flux Q , sink flux S , and content M .

- The residence time is the time spent in a reservoir by an individual atom or molecule. It is also the age of the molecule when it leaves the system. If the pathway of a tracer from the source to the sink is characterized by a physical transport then the word transit time can also be used.
- Even for a single chemical substance the different atoms or molecules within a mol of that substance will have different residence time. Hence we need to deal with a probability density function of residence times $\varphi(\tau)$ where $\varphi(\tau) d\tau$ describes the fraction of the tracer having a residence time in the interval τ to $\tau + d\tau$.

The average residence time is defined by:

$$\tau_r = \int_0^{\infty} \tau \varphi(\tau) d\tau$$

The probability density distribution can have many shapes



- Example A corresponds to a lake with inlet and outlets on opposite sites of the lake. Most water molecules will have a residence time roughly equal to the time it takes for the mean current to carry water from the inlet to the outlet.
- Example B illustrates exponential decay. A simple example could be the reservoir of all ^{238}U on Earth. The half-life of this radionuclide is 4.5×10^9 years. Since this roughly corresponds to the age of the Earth the ^{238}U content of Earth is now half compared to what it was at the time of its formation.
- Example C corresponds to a case where the sink is close to the source and the youngest particles have faster removal rates.

FIG. 4-4 The age frequency function $\psi(\tau)$ and the residence time frequency function $\phi(\tau)$ and the corresponding average values τ_a and τ_r for the three cases described in the text: (a) $\tau_a > \tau_r$; (b) $\tau_a = \tau_r$; (c) $\tau_a > \tau_r$. (Adapted from [Bolin and Rodhe \(1973\)](#)

with permission from the Swedish Geophysical Society.)

The age

- The age of an atom or molecule in a reservoir is the time since it entered the reservoir. As with the residence time, the probability distribution $\psi(\tau)$ of ages has a shape that depends on the situation.
- In a steady-state reservoir, however, $\psi(\tau)$ is always a non-increasing function
- The average age of atoms in a reservoir is given by:

$$\tau_r = \int_0^{\infty} \tau \psi(\tau) d\tau$$

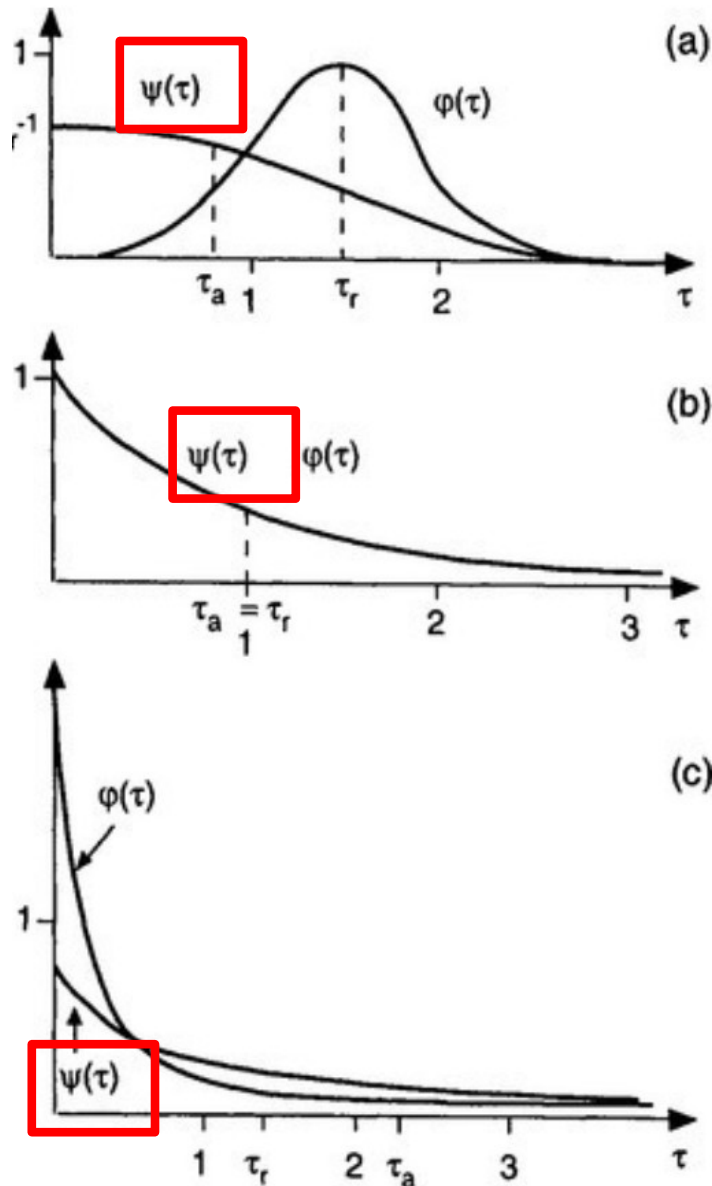


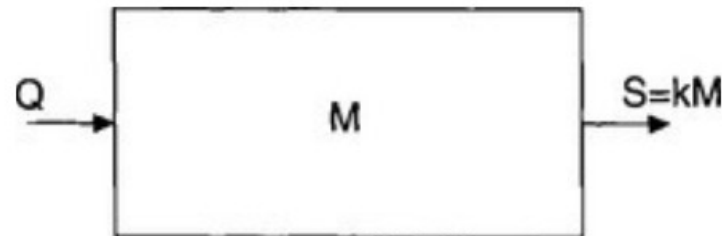
FIG. 4-4 The age frequency function $\psi(\tau)$ and the residence time frequency function $\phi(\tau)$ and the corresponding average values τ_a and τ_r for the three cases described in the text: (a) $\tau_a > \tau_r$; (b) $\tau_a = \tau_r$; (c) $\tau_a < \tau_r$. (Adapted from [Bolin and Rodhe \(1973\)](#) with permission from the Swedish Geophysical Society.)

Relations between τ_0 , τ_r and τ_a

- It can be shown that for a reservoir in steady-state τ_0 is equal to τ_r i.e. the turnover time is equal to the average residence time spent in the reservoir by individual particles.
- This is useful. For example if τ_0 is known from budget considerations (e.g. the water flow at the inlet and outlet) and if the average transport velocity (V) (e.g. the speed of the current in the lake) is known, the average distance ($L=V\tau_0$) over which the transport takes place in the reservoir can be estimated.
- The relationship between τ_0 and τ_a is not that simple as τ_a can be smaller or bigger than τ_0 . For a well mixed reservoir with a first-order removal process $\tau_a = \tau_0$ typically applies.
With respect to the human population τ_a is only approximately half of τ_0 . In India $\tau_a = 28$ years (average age) while the average residence time τ_0 (life expectancy at birth or average age at the time of death) = 72 years.
- In situations where most atoms leave the reservoir very soon and few remain very long $\tau_a > \tau_0$. This represents the short circuit case where e.g. the inlet and outlet of a lake are very close together.

Response time

- The response time (also known as relaxation time or adjustment time) of a reservoir is the time scale that characterizes the adjustment to equilibrium after a sudden change in the system



- Consider a reservoir initially in steady-state with $Q_0 = S_0$ and M_0 with a first order removal process $S_0 = kM_0$. The turnover time of this reservoir is:

$$\tau_0 = \frac{M_0}{S_0} = \frac{M_0}{kM_0} = \frac{1}{k}$$

- Suppose now that the source strength is suddenly changed to a new value Q_1 . The adjustment process is described by a differential equation with the initial conditions $M(t=0) = M_0$

$$\frac{dM}{dt} = Q_1 - S_t = Q_1 - kM_t$$

- The solution $M(t) = M_1 - (M_1 - M_0)e^{(-kt)}$ approaches a new steady state value

of $M_1 = \frac{Q_1}{k}$ with a response time of $\tau_0 = \frac{1}{k}$

Response time

- The solution $M(t) = M_1 - (M_1 - M_0)e^{(-kt)}$ approaches a new steady state value of $M_1 = \frac{Q_1}{k}$ with a response time of $\tau_0 = \frac{1}{k}$
- The change in the reservoir mass from the initial to the final condition is shown in the figure below.

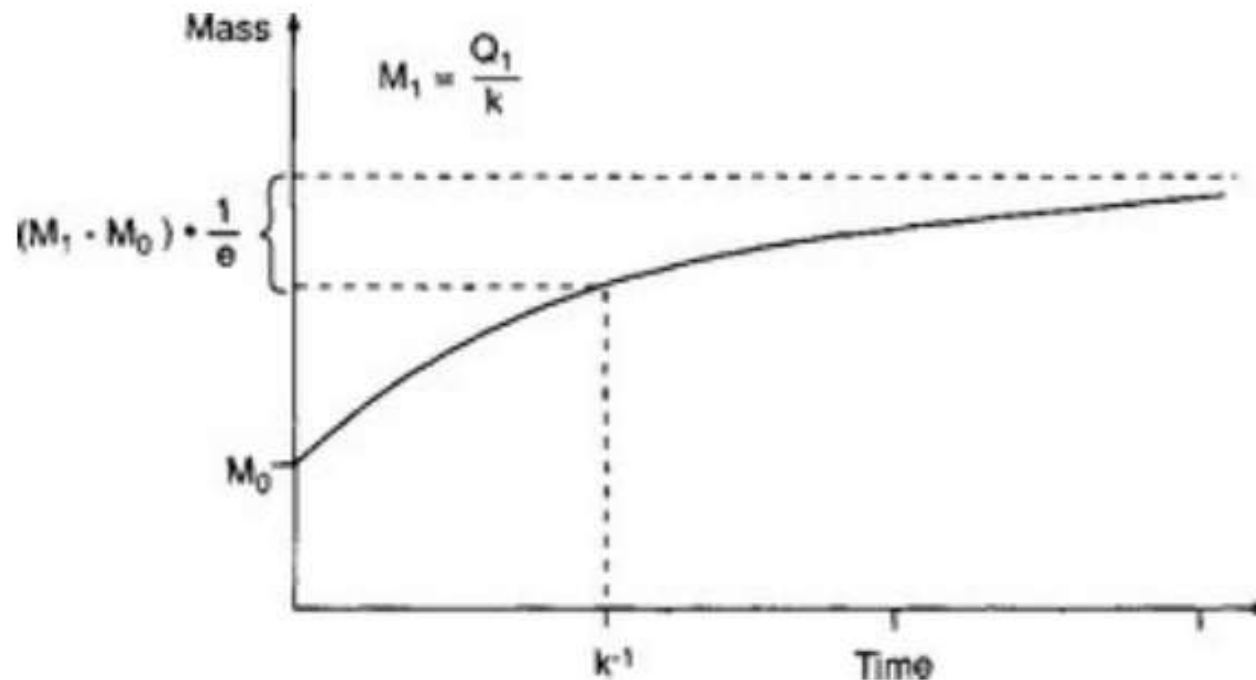
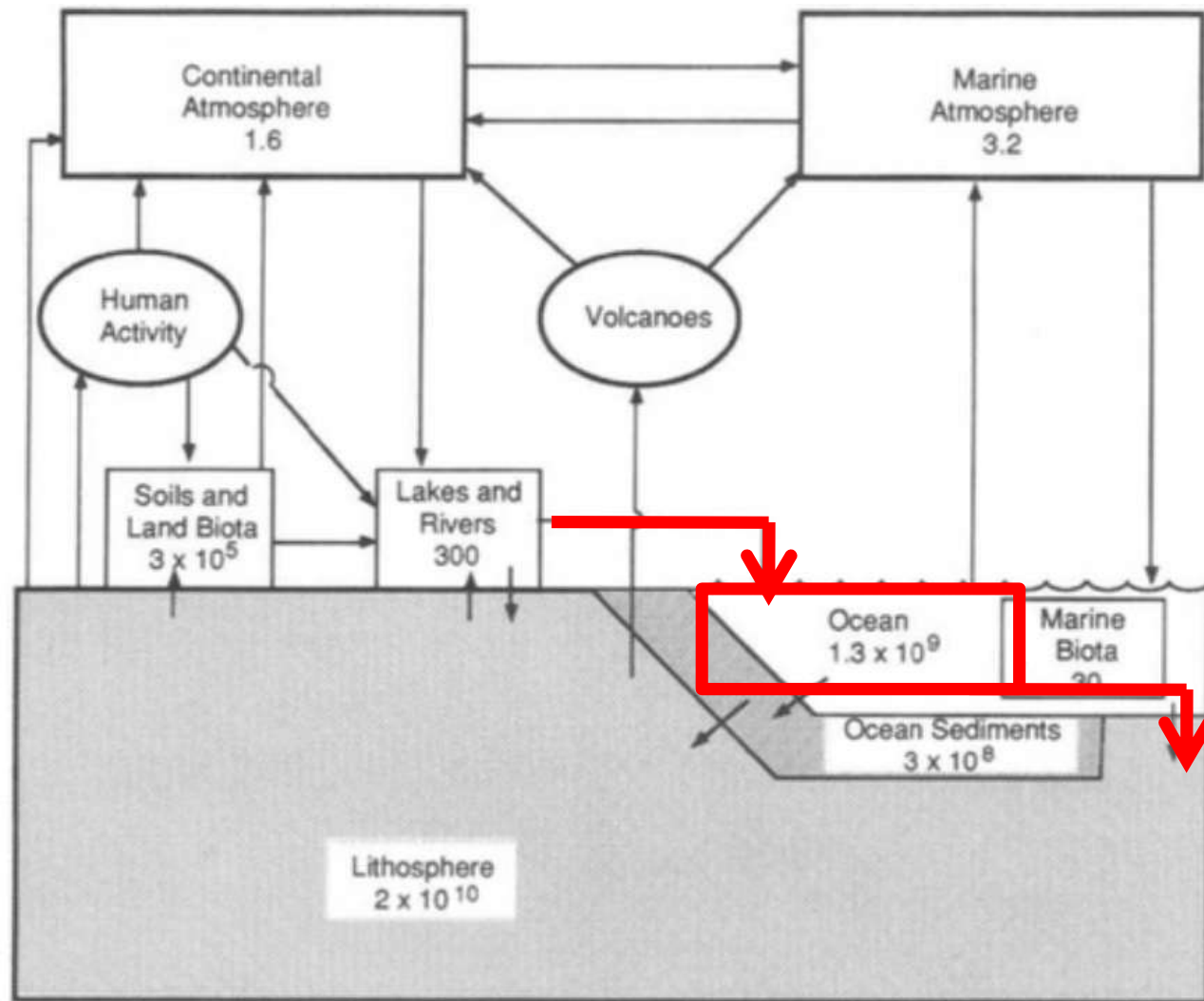


FIG. 4-5 Illustration of an exponential adjustment process. In this case, the response time is equal to k^{-1} .

Example of a response time calculation



Let us consider just one of the reservoirs of the carbon cycle:

The ocean contains 1.3×10^9 Tg S as burden.

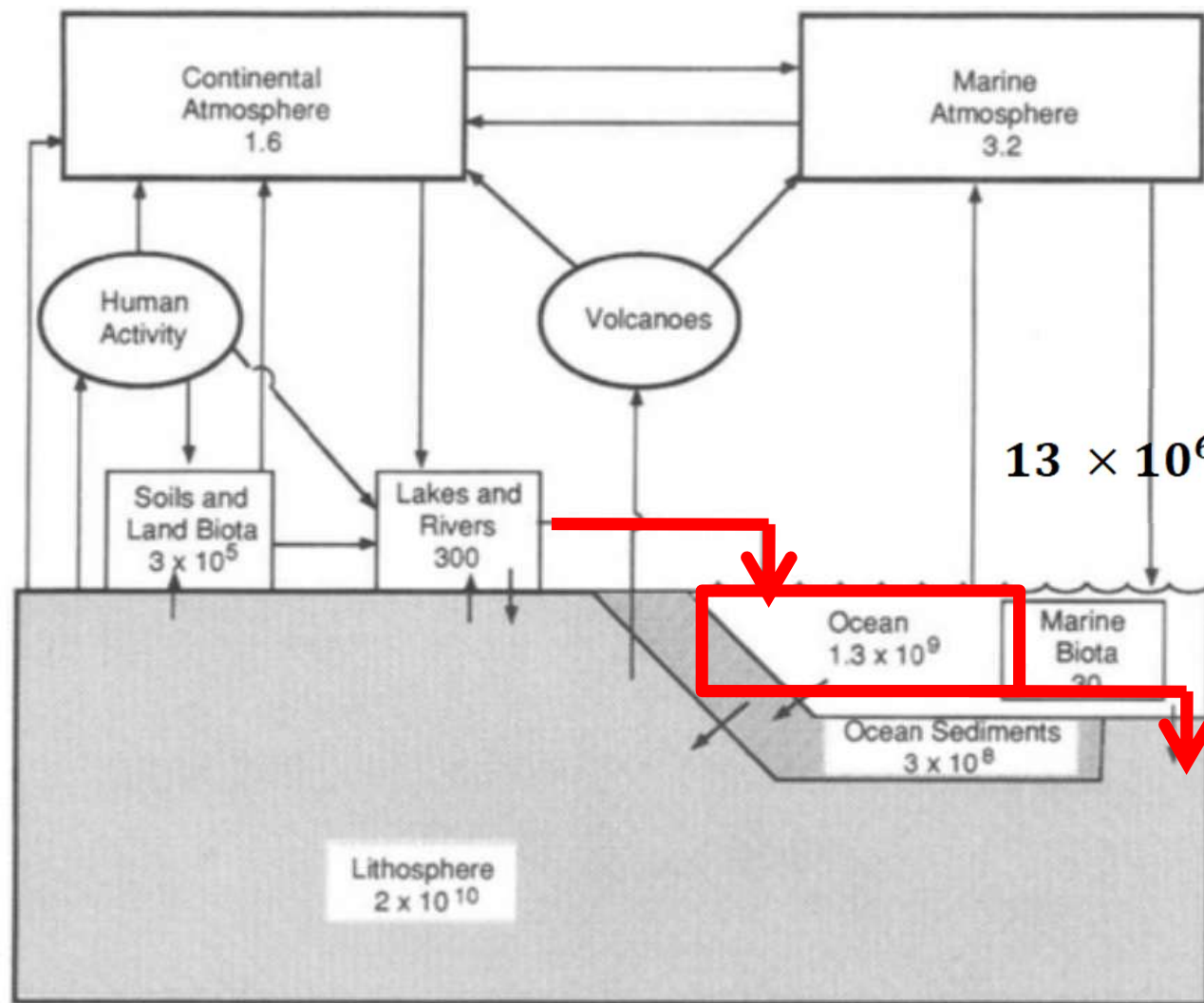
In preindustrial time material used to get added through river runoff at a rate of 100 Tg S yr^{-1} (Q_0)

The main sink of 100 Tg S yr^{-1} ($S_0 = kM_0$) was incorporation into the lithosphere via hydrothermal circulation at mid-ocean ridges

If we make the unrealistic assumption that the present day increased runoff of 200 Tg S yr^{-1} (Q_1) would continue indefinitely, then how fast would the sulfur burden of the ocean adjust to a new equilibrium?

FIG. 13-1 Major reservoirs and burdens of sulfur. Units are Tg (10^{12} g) S.

Breaking the problem into steps



Let us start with calculating τ_0

$$\tau_0 = \frac{M_0}{S_0} = \frac{M_0}{kM_0} = \frac{1}{k}$$

The time scale characterizing the adjustment is

$$13 \times 10^6 [\text{yr}] = \frac{1.3 \times 10^9 [\text{Tg}]}{100 [\text{Tg yr}^{-1}]} = \frac{M_0}{kM_0} = \frac{1}{k}$$

Now let us use the last part of the equation to calculate the proportionality constant k , which doesn't change.

$$\frac{1}{13 \times 10^6 [\text{yr}]} = 7.7 \times 10^{-8} [\text{yr}^{-1}]$$

... and use $M_1 = \frac{Q_1}{k}$ to calculate M_1

...which is double of M_0

$$M_1 = \frac{200 [\text{Tg yr}^{-1}]}{7.7 \times 10^{-8} [\text{yr}^{-1}]} = 200 \times 13 \times 10^6 [\text{Tg}] = 2.6 \times 10^9 [\text{Tg}]$$

FIG. 13-1 Major reservoirs and burdens of sulfur. Units are Tg (10^{12} g) S.

Reservoirs in Non-steady state

- Let us analyse the situation when one observes a change in the reservoir content and wants to draw conclusions regarding sources and sinks. Let us re-write this equation:

$$\frac{dM}{dt} = Q_1 - S_t = Q_1 - kM_t$$

- In the following format by dividing everything by M

$$\frac{1}{M} \frac{dM}{dt} = \frac{Q}{M} - \frac{1}{\tau_0} \quad \text{and applying} \quad \tau_0 = \frac{1}{k}$$

- Now let us denote the left side of the equation, the observed rate of change in the reservoir content as τ_{obs}^{-1}
- If the mass were to increase by 1% per year then τ_{obs} would be 100 years.
- Two limiting cases can be singled out $\tau_{obs} \gg \tau_0$ and $\tau_{obs} \ll \tau_0$.

Reservoirs in Non-steady state

with $\tau_{obs} \gg \tau_0$

- With a 1% change per annum in M if $\tau_{obs} \gg \tau_0$ there has to be an approximate balance between the two terms on the right-hand side of the equation:

$$\tau_{obs}^{-1} = \frac{1}{100} = \frac{Q}{M} - \frac{1}{\tau_0}$$

$$\frac{Q}{M} \approx \frac{1}{\tau_0} \quad or \quad Q \approx \frac{M}{\tau_0}$$

- This means that the observed change in M mainly reflects a change in the source flux Q or the sink function.

Example calculation for Reservoirs in Non-steady state with $\tau_{obs} \gg \tau_0$

- The atmospheric methane concentration has been increasing by about 0.5% per annum in recent years and the turnover time with respect to reaction with the OH radical is estimated to be roughly 10 years.

$$\tau_{obs}^{-1} = \frac{1}{100} = \frac{Q}{M} - \frac{1}{\tau_0} \qquad \frac{Q}{M} \approx \frac{1}{\tau_0} \quad or \quad Q \approx \frac{M}{\tau_0}$$

- Step 1: Check if you are calculating with the right equation.** Rate of change 0.5% p.a.; Convert % to ratio: 0.5% = 0.005; calculate 1/0.005=200; see if 200>>10.
- Step 2: Estimate if source or sink is changing.** In this case the anthropogenic sources (cattle farming, rice farming, fossil fuel extraction, bad waste & waste water management) are changing faster than the sink.
- Step 3: Approximate that the source strength must be changing at the same rate as the reservoir content is increasing**

Example calculation for Reservoirs in Non-steady state with $\tau_{\text{obs}} \ll \tau_0$

- In this case $\frac{dM}{dt} \approx Q$
- An example would be the increase in sulfur hexafluoride SF_6 concentration in the atmosphere by 0.5% p.a. due to anthropogenic emissions. This gas is chemically stable and has a turnover time of 3000 years.
- **Step 1: Check if you are calculating with the right equation. Rate of change 0.5% p.a.; Convert % to ratio: $0.5\% = 0.005$; calculate $1/0.005=200$; see if $200 \ll 3000$.**
- **Step 2: In this case the source strength is not changing but rather the sink is so close to non-existent (so slow) that a source with constant source strength drives up the burden by the proportionate amount.**

PRACTICE QUESTIONS

Please go through them tonight so that we can discuss where you get stuck tomorrow or latest before Friday's class

Long answer style & numericals

1. Use figure 4.3 to determine the turnover time of long lived biota on land and explain which process dominates the turnover time of the biomass.
2. Consider a reservoir with two separate sources Q_1 and Q_2 and a single sink S . The magnitudes of Q_1 and S and their uncertainties have been estimated to be 75 ± 20 and 100 ± 30 (arbitrary units).
 - a) In a situation where there is no direct way of measuring Q_2 how would you derive the magnitude and uncertainty of the flux Q_2 ?
 - b) Which fundamental assumption about the reservoir do you need to make to estimate Q_2 ?
3. What is the relationship between turnover time the average transit time and the average age in a reservoir where all particles spend exactly equal time in the reservoir?

Long answer style & numericals

1. Consider a reservoir with a source flux A and two sink fluxes S_1 and S_2 . S_1 and S_2 are proportional to M with the proportionality constants k_1 and k_2 of 1 yr^{-1} and 0.2 yr^{-1} . The system is initially in steady state with $M=M_0$ and $Q_0=S_{1,0}+S_{2,0}$. Describe the response time if the source is suddenly reduced to half its initial value and calculate the new equilibrium concentration M_1 obtained if the new source flow Q_1 persists.
2. Consider the water balance of a lake with a constant source flux Q . The outlet is the “threshold” type where the sink is proportional to the mass of water above a threshold value M_1 .
 $S_0=k(M_0-M_1)$. Calculate the turnover time of water at steady state for Q that is sufficient to keep $M_0>M_1$ and the response time relative to a change in Q .
3. Both the Atmosphere is currently not in equilibrium with respect to their carbon content. Based on figure 4.3 and your understanding of non-equilibrium systems, what can you conclude about the current status of sources and sinks. Would it change your consideration if you look at “the land” and “the ocean” as net C sink $S_{\text{net}}=(S-Q)$ rather than gross sink or not? Explain!